

ALEXANDER BAY, South Africa

WMO: 684030

Lat: 28.570S

Lon: 16.529E

Elev: 28

StdP: 100.99

Time Zone: 2.00 (E02)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	6.1	7.2	-4.9	2.5	21.9	-2.2	3.2	21.0	12.7	15.9	11.4	16.2	1.5	150	0.476

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	7.0	32.0	17.1	29.1	16.3	26.6	16.2	20.6	25.8	19.7	24.0	19.1	23.1	5.2	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.9	13.8	22.1	18.2	13.1	21.4	17.6	12.7	21.0	59.3	25.8	56.5	24.0	54.5	23.1	27.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
14.2	12.3	10.5	DB	4.4	37.1	0.9	2.3	3.8	38.7	3.3	40.0	2.8	41.3	2.2	43.0	(4)
			WB	3.1	21.9	0.7	1.2	2.5	22.8	2.1	23.5	1.7	24.1	1.1	25.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	17.2	19.9	19.9	19.0	18.1	16.5	15.2	15.0	14.4	15.2	16.4	17.8	19.3	
	DBStd	3.19	1.62	1.73	2.28	3.40	3.03	3.16	3.61	2.62	2.50	1.99	1.68	1.55	
	HDD10.0	1	0	0	0	0	0	0	1	0	0	0	0	0	
	HDD18.3	699	3	3	16	41	76	106	122	131	101	65	28	8	
	CDD10.0	2634	307	277	279	244	201	157	154	136	157	199	234	288	
	CDD18.3	292	52	47	36	35	18	14	17	8	8	7	12	38	
	CDH23.3	1536	94	96	171	309	178	132	163	95	122	72	59	45	
	CDH26.7	562	15	18	68	146	74	39	48	37	57	29	24	6	
Wind	WSAvg	4.0	4.9	4.5	4.1	3.3	2.9	3.1	3.1	3.5	4.1	4.7	5.0	5.2	
Precipitation	PrecAvg	42	1	2	2	4	4	8	5	6	4	4	1	2	
	PrecMax	91	10	13	15	18	37	47	21	30	26	18	12	16	
	PrecMin	8	0	0	0	0	0	0	0	0	0	0	0	0	
	PrecStd	20	2	3	3	5	8	10	5	7	6	4	3	3	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	27.9	29.3	33.2	35.9	32.9	30.6	30.7	31.2	33.0	31.4	30.0	26.6	
		MCWB	18.4	20.6	19.2	19.3	16.6	15.6	15.0	14.5	17.9	16.8	16.2	17.8	
	2%	DB	25.2	25.4	27.5	30.9	28.5	27.1	27.8	25.8	26.2	25.1	24.1	24.3	
		MCWB	19.0	19.1	18.8	18.3	15.3	14.3	14.2	12.8	15.1	16.0	16.2	18.2	
	5%	DB	24.0	24.2	24.2	26.1	23.9	23.6	24.5	21.1	21.6	21.7	22.2	23.2	
		MCWB	18.6	18.8	18.3	17.1	14.3	13.4	13.3	11.9	13.8	15.5	16.4	17.8	
	10%	DB	23.0	23.1	22.6	22.8	21.1	20.5	21.1	18.5	19.1	20.1	21.1	22.4	
		MCWB	18.0	18.3	17.6	16.6	14.8	12.9	12.3	12.7	13.7	15.1	16.1	17.5	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	21.1	21.4	21.4	21.1	18.4	17.8	16.6	15.6	18.7	19.3	18.5	20.2
			MCDB	24.8	26.7	29.5	31.2	24.5	26.0	26.3	23.6	30.3	28.7	23.1	23.2
2%		WB	20.1	20.1	19.9	19.6	16.9	15.8	15.2	14.5	15.6	17.1	17.6	19.2	
		MCDB	23.4	23.8	24.9	27.0	22.3	21.8	23.2	20.3	22.5	21.7	21.4	22.4	
5%		WB	19.4	19.5	18.9	18.2	16.2	14.9	14.3	13.9	14.8	16.2	17.1	18.6	
		MCDB	22.6	22.9	22.9	23.7	20.8	20.2	20.8	18.3	19.6	20.0	20.7	21.8	
10%		WB	18.7	18.8	18.1	17.2	15.5	14.1	13.5	13.3	14.2	15.5	16.6	18.0	
		MCDB	21.8	22.2	21.7	21.8	19.7	18.7	18.7	17.2	18.1	19.1	20.1	21.3	
Mean Daily Temperature Range	5% DB	MDBR	7.0	7.1	8.0	9.5	9.7	10.5	10.7	9.7	9.2	8.2	7.7	7.0	
		MCDBR	8.0	8.2	11.4	16.3	16.1	16.2	16.5	17.1	16.1	12.3	9.6	7.8	
		MCWBR	3.3	3.7	4.8	6.1	6.7	6.7	6.8	7.2	6.5	5.4	4.2	3.3	
	5% WB	MCDBR	7.0	7.2	9.5	12.2	10.6	11.6	12.4	10.5	10.7	9.1	8.0	6.7	
		MCWBR	3.4	3.4	4.3	5.3	5.4	6.1	6.4	5.5	5.2	4.5	4.0	3.2	
Clear-Sky Solar Irradiance	taub	0.360	0.370	0.355	0.328	0.308	0.298	0.308	0.328	0.354	0.361	0.342	0.346		
	taud	2.464	2.461	2.487	2.549	2.569	2.556	2.518	2.449	2.358	2.368	2.470	2.475		
	Ebn at Noon	981	954	937	916	890	877	880	900	922	951	993	997		
	Edh at Noon	119	116	107	91	81	77	83	99	120	126	118	119		
All-Sky Solar Radiation	RadAvg	8.10	7.40	6.39	5.03	3.94	3.47	3.72	4.52	5.79	7.06	8.10	8.51		
	RadStd	0.34	0.27	0.20	0.19	0.13	0.11	0.15	0.18	0.21	0.18	0.24	0.30		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (4 neighbors)	+0.44	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page